

Zhe Chen 陈哲

zc2325@ic.ac.uk / ssyzc1011@gmail.com
github.com/CZ114 | Personal website



Education

Imperial College London MRes, Medical Robotics Research completed; formal degree award pending	2025.10 – 2026.09
University of Nottingham Ningbo China BEng (Hons), Electrical and Electronic Engineering First-Class Honours	2021.09 – 2025.06

Research and Projects

Multimodal AI Analysis Platform for Alzheimer's Disease Research Imperial College London Hamlyn Centre · I-thread Lab MRes Research	2026.01 – 2026.09
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Addressed fragmented multimodal records and difficult evidence review in longitudinal monitoring. Responsible for **wearable sensing, the data platform, and agent framework development and evaluation**, building a research prototype from signal acquisition to auditable analytical reports.

- **Acquisition and quality control:** Integrated ESP32-S3 PPG, GSR, IMU and audio channels. Detected no frame loss across **27 serial recordings and 136,562 frames**, with stable delivery at approximately **50 Hz**, providing continuous data for subsequent analysis.
- **Signal algorithms:** Combined spectral analysis, autocorrelation and motion suppression for heart-rate estimation. Achieved **MAE of 5.57 bpm** in leave-one-subject-out (LOSO) validation on 15 PPG-DaLiA participants (device baseline: 11.84 bpm); marked motion-corrupted heart-rate features as unusable.
- **Agent analysis workflow:** Designed deterministic tools, retrieval, memory and bounded workflows. In a fixed-model ablation, the full configuration increased movement classification accuracy from **35.7% with a bare prompt to 81.4%** ($p = 0.0007$), reducing parsing failures from **15 to 0**.
- **End-to-end comparison:** Evaluated **70 recordings across 5 movement classes** from 6 healthy volunteers using LOSO and the same 18 features. The best agent achieved **82.9% accuracy** and **0.804 macro-F1**, versus 78.6% / 0.713 for random forest, supporting feasibility on a small dataset.
- **Auditable outputs:** **66/70 (94.3%)** outputs passed all traceability checks. Retained recording identifiers, evidence, quality limitations and tool execution states so reviewers could inspect the basis of the analysis, not just its predicted class.
- **Evaluator calibration:** Matched the official BFCL evaluator's decisions on **400/400 single-turn cases**; both scored the same outputs at **91.25%**. Calibrated the in-house evaluator to distinguish scoring differences from model capability.
- **Retrieval evaluation:** Implemented hybrid retrieval and HyDE comparisons. Achieved **Recall@10 = 0.846** and **MRR = 0.753–0.774** on a 24-question internal set, assessing context provision through retrieval coverage and ranking quality.
- **Simulation benchmarks:** Reduced mean dialogue turns for all six models in AgentClinic comparisons. Two model conditions improved task accuracy from **53% / 77% to 67% / 87%**, assessing the framework's effect on simulated task efficiency and performance.

Research and Projects

Speech Evidence & Clinician Report Chat

2026

Imperial College London · Hamlyn Centre · I-thread Lab | Research Support
Manuscript in preparation · Target journal: Nature Medicine

- Assisted with checking and verifying speech metrics within the collaborative research project.
- Helped organise the report page and its existing content for review.
- Implemented part of the chat functionality within the report interface.

Reinforcement Learning for Port Container Relocation

2024.08 – 2025.06

University of Nottingham Ningbo China | Undergraduate Dissertation

- Built a three-dimensional container-yard simulator to reduce unnecessary handling; designed state representations and reward functions, and implemented A2C/DQN training and evaluation pipelines.
- Against a **random-action baseline**, DQN reduced rehandles by approximately **23% in simple scenarios**, while A2C reduced handling moves by approximately **40% in complex scenarios**, demonstrating scope for learning-based scheduling improvements.

Research Experience

Imperial College London · Hamlyn Centre · I-thread Lab

Current

Research Assistant (RA) | AD-related wearable sensing, AI agents and speech research

UNNC Smart Medicine Laboratory

2023.09 – 2025.07

Research Intern / Project Lead | Smart-Home IoT and Federated Learning for Sleep Apnoea

- Addressed the lack of expert labels for home heart-rate data. Built an M5Stack/ESP32 + MQTT gateway, configured FATE/PyTorch experiments, and ran **confidence-thresholded pseudo-labelling and four federated training rounds**, comparing pseudo-labels with a control treating home samples as normal.
- Achieved **81% external-test accuracy in round four versus 76% for the control (+5 percentage points)**; sleep-apnoea class precision / recall were **0.93 / 0.78**, supporting the feasibility of improving models with unlabelled home data.
- Contributed to drafting a collaborative manuscript as a listed co-author, providing experimental evidence for collaborative modelling with home data lacking expert annotations.

Open-Source Tools

Vibedog-for-agents 2026.05 – 2026.06 | AI-assisted development

Agent activity display, human approval and a stop control. github.com/CZ114/Vibedog-for-agents

esp32-audio-ai | Audio acquisition and transcription precursor to the main project

ESP32 audio capture, Whisper transcription, and pause / speech-rate statistics.

github.com/CZ114/esp32-audio-ai

Technical Skills

Agents and Evaluation: Function Calling, RAG, workflows and memory; BFCL calibration, retrieval evaluation, LOSO, ablation studies and evidence traceability.

Algorithm Foundations: Python, PyTorch, A2C/DQN, federated learning and pseudo-labelling; systematic study of Transformer/Attention and the principles of SFT, RLHF and PPO.

Robotics and Embedded Systems: ESP32-S3, I2S, BLE, UDP, MQTT, CoppeliaSim, robot kinematics, Ubuntu, Jetson Nano and Altium Designer.